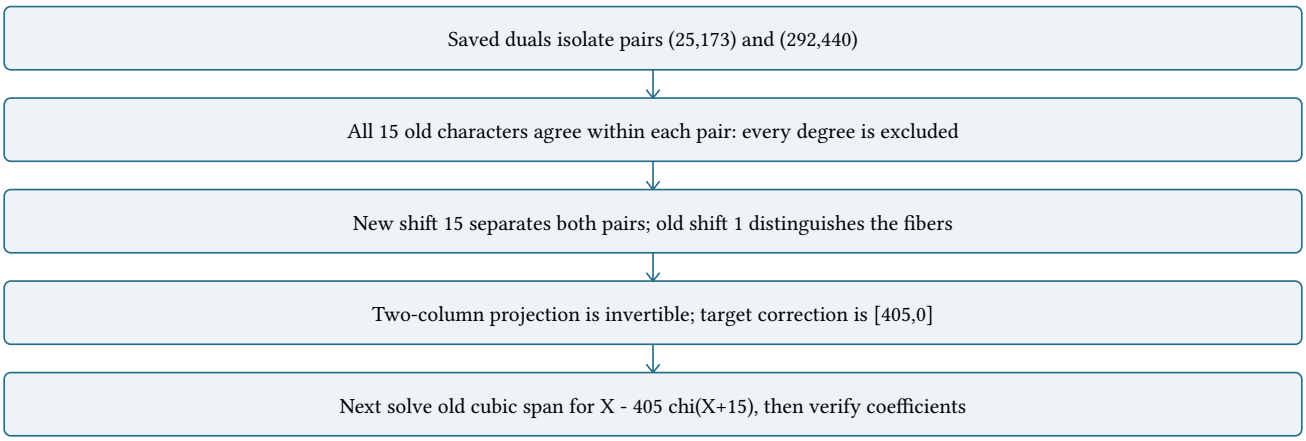
Character fibers and targeted span repair

2026-11-03 · Derived theorem / exact checks · No performance measurement

Stronger obstruction. The saved F479 width-15 duals combine to give 178(e25−e173) and 178(e292−e440), each normalized against X. Independent square-set evaluation confirms all15 old character values agree within each pair. Thus no function of those characters can represent X: increasing degree alone cannot repair this specific width.



Targeted repair. Let U0=chi(X+15), U1=chi(X+1)chi(X+15). Pairwise projection gives M=[[2,−2],[2,2]], determinant8, and target pairings[331,331]. The checked correction is [405,0]. Therefore X−405chi(X+15) has zero pairing with both old difference functionals.

Rank premise. The preserved frozen study reports old cubic feature rank477. The two pair functionals then span the entire left kernel, so the corrected RHS belongs to the old cubic span. This proves existence of a representation using old width-15 cubic features plus one new linear character. Its old-feature coefficients have not yet been constructed. The full old rank is not replayed here; a Gram inverse cannot be reused without establishing image equality.

Current exact checks	Result / scope
Saved dual combinations	All479 entries checked
Old pair singleton equalities	30, independently checked
New quotient matrix	Rank2; residual pairings0,0
Local software suite	5 methods passed, including subcases/mutations
Historical rank premise	477 from prior frozen computation

Software cases also exercise F7 scalar/pair Schur updates and counters, normalized witness lifts, invalid pivots and frame order, streaming dependencies, finite caps, and corrupted witnesses/theorem records. Prior static records stay unchanged. Root-pivot integration still lacks numerical instance validation.

Costs. Construction: 4q+4 character evaluations, at most q distinguishing offset tests, O(q) storage, then ten field products, seven subtractions and one inversion. Native pow is unmetered. The small-prime reference checker uses O(r) square-set work. A consistent old-span solve remains the next major step. No total-time or scalar-recovery improvement is measured.

Custody / next. Sources fixed at5ed039e before the bounded local unittest run: one CPU,512MiB,30seconds,one attempt; exit0. Certificate packet separately materialized and independently checked. No cloud job, build or budget change. Next construct and verify old-feature coefficients for the corrected RHS, then admit a full-cost compiler comparison with all historical controls.

Sources. Saved feature-span-dual certificate/rank summary; NIST DLMF27.9; repair-certificate.json, certificate-check.json and unit-check-receipt.json.